

Circulatory System Simulation

Interactive Learning Simulation for Children

1. Project Title:

Circulatory System Simulation

2. Description of the Project

The **Circulatory System Simulation** is an interactive web-based educational application designed to help users understand the structure and functioning of the human circulatory system. The project visually demonstrates how blood circulates throughout the body, transporting oxygen, nutrients, and waste materials between organs and tissues.

The simulation provides an engaging learning experience through animations, diagrams, and user interactions, making complex biological concepts easier to understand for students and healthcare enthusiasts.

3. Problem Statement

Understanding the human circulatory system through traditional textbooks and static diagrams can be difficult for students. Many learners struggle to visualize how blood flows through the body and how the heart, arteries, veins, and capillaries work together. There is a need for an interactive and engaging learning tool that simplifies these concepts and improves students' understanding of the circulatory system through visual simulations and real-time interaction.

4. Solution

This project provides an interactive simulation of the human circulatory system. Students can learn how blood flows through the body and understand the functions of the heart, arteries, veins, and capillaries through visual animations and information panels.

Heart Simulation

Students observe how the heart pumps blood throughout the body.

Blood Flow Simulation

Students learn the movement of oxygen-rich and oxygen-poor blood through blood vessels.

Interactive Learning

Users can click on different parts of the circulatory system to view detailed information and improve their understanding.

5. Features

Educational Features

- Interactive visualization of the human circulatory system.
- Demonstrates blood flow throughout the body.
- Explains the functions of the heart, arteries, veins, and capillaries.
- Click-to-view information panels for better understanding.
- Simple and user-friendly interface for effective learning.

Technical Features

- Built using HTML, CSS, and JavaScript.
- Interactive and responsive design.
- Easy-to-use interface.

Learning Outcomes

- Understand blood circulation in the human body.
- Learn the functions of the heart and blood vessels.
- Improve knowledge of the circulatory system through visual learning.

6. Tech Stack Used

Technology	Purpose
HTML5	Structure and content of the website
CSS3	Styling, layout, and animations
JavaScript	Interactive functionality and logic
Emojis & Icons	Visual representation of experiments
Responsive Design	Compatibility across devices
Vercel	Hosting, deployment, and global content delivery

The project was developed using front-end web technologies

7. How to Run the Project

Follow these steps to run the Circulatory System Simulation:

Step 1

Download or clone the project files.

Step 2

Ensure all project files are stored in the same folder.

Step 3

Open the project folder.

Step 4

Locate the **index.html** file.

Step 5

Double-click the file or open it using any modern web browser such as:

- Google Chrome
- Microsoft Edge
- Mozilla Firefox

Step 6

The Circulatory System Simulation will load automatically.

Step 7

Explore the circulatory system by interacting with the different components and information panels.

Step 8

Observe blood circulation and learn the functions of the heart, arteries, veins, and capillaries.

8. Deployment Link (Deployed in vercel):

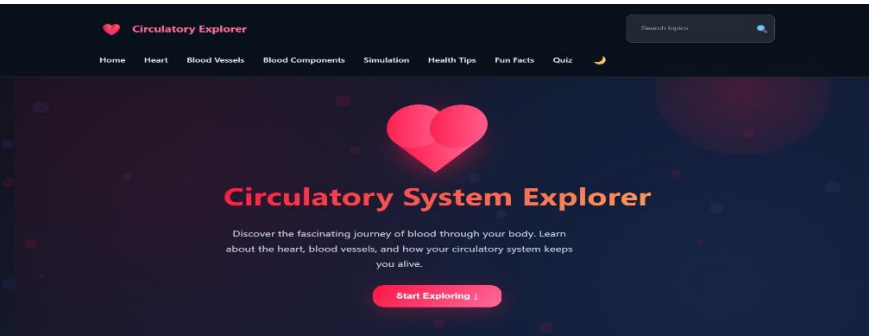
<https://heat-quest-adventure.vercel.app/experiment.html>

9. Team Members

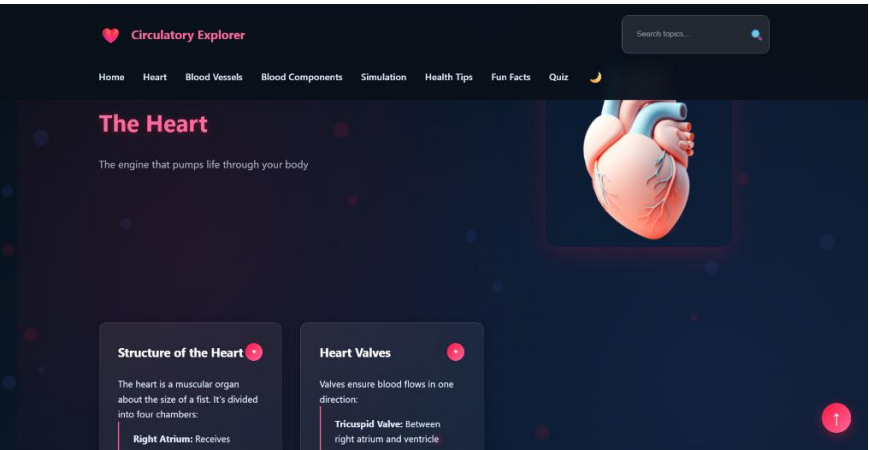
#	Team Member(s)	Role / Responsibility	Task Assigned
1	☐ Krishna Sri	Project Planner & Requirements Analyst	Storyboard creation, project requirement gathering and analysis
2	☐ Rahimunnisa ☐ Aditya	Simulation Developers	Development of simulation modules using HTML, CSS, JavaScript, and Index.html integration
3	☐ Harsha ☐ Avinash	Multimedia & Design Team	Creation and management of project images, graphics, and visual assets
4	☐ Sarika ☐ Roopa	Content Management Team	Preparation of About Team documentation and project content files
5	☐ Aditya ☐ Roopa	Deployment Engineers	Application deployment, hosting configuration, and deployment management
6	☐ Naga Krishna ☐ Vardhan	Documentation Specialists	Preparation of project documentation, code explanation, theoretical concepts, and inclusion of deployment link in the final report

10.Website Screenshots

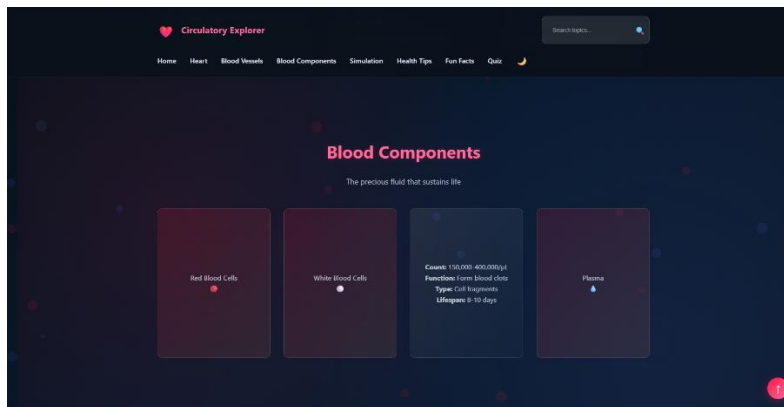
Home Page :



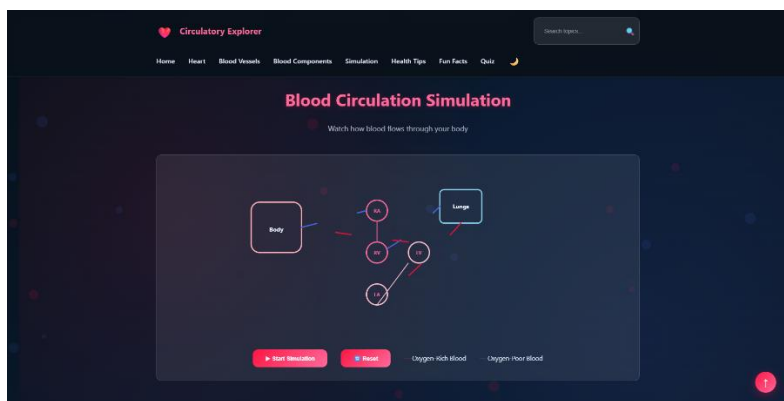
Heart Simulation



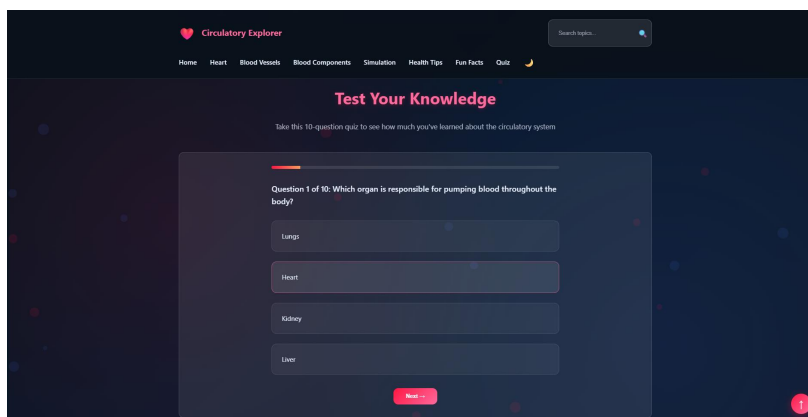
Blood Components



Blood Circulation Simulation



Test your knowledge (Quiz)



11. Conclusion

The Circulatory System Simulation effectively bridges the gap between theoretical learning and practical visualization. By integrating interactive features and educational content, the project serves as a valuable tool for students, educators, and anyone interested in learning about the human circulatory system.